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Semiconductors | North America

Our shopping list given recent selloff

We acknowledge that with recent risks our group may be more rangebound - but we are buyers as we approach the low end of the range. Top preference remains NVDA/AVGO but increasingly looking to play offense with MU/SNDK

Key Takeaways

- There is some risk Incrementally, partly from the midterms but more so from stock reactions - fear of financial contagion is significant
- Those threats may continue to weigh on stocks but in a rangebound group we want to be more optimistic at the low end of the range
- We force-ranked sub sector exposures by preference 1) GPU/XPU compute, 2) memory, 3) SPE, 4) analog, 5) networking, 6) CPU
- Top Pick is NVDA but risk reward looking more favorable for MU/SNDK as well.

We would acknowledge some risk from recent events, but still see business visibility as exceptionally strong. Calling an absolute bottom is difficult, and we aren't trying to do that today - but we do think that the risk reward is significantly improving across our coverage as the selloff seems overdone.

Before we list stocks that we like, some thoughts on the current state of the AI buildout.

Our biggest fear is fear itself. Business is exceptionally strong across the board, and talking with spenders - semiconductor customers - it seems likely to be durable. The desire to spend for multiple years, from well capitalized companies, is very much intact. Even neoclouds plagued with high unsecured interest rates are having trouble getting all of the compute that they need into the space that they already have. But the capex is funded by capital markets, and the contagion across asset classes is notable and worth watching. In particular, an equity market that is less thrilled by capex increases could start to impact willingness to spend. This was the biggest risk post Liberation Day, the biggest risk around Middle Eastern conflicts, and is the biggest risk today.

The notion that "it's all about 2028 now" is fair but not damning in itself - this feels no different than the last 3 years. At some point, when investors become comfortable with the next 12 months, the debate becomes months 13 through 24, which is always difficult for semiconductor investors to get comfortable with given lead times of at most 12 months, As investors in late 2023 became comfortable with CY24, stocks sold off because "its all about 2025".

Of course, one of these years that anxiety will be correct. But currently the amount

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North America
Industry View **Attractive**

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of unmet demand in 2027 as of right now - acknowledging that it could change - certainly will push enough into 2028 that the 2028 outlook feels as good as it could feel this long in advance. And the fact that there are bottlenecks - very much including land power shell - is not the threat that bears make it out to be; it is in fact the reason that the growth has been so consistent and durable.

But aren't there new barriers to land power shell (LPS) post the mid terms? Yes, and we work closely with Stephen Byrd and the power team to assess that. But as we have talked to buyers over the last week, the sense is that LPS is still not the biggest challenge to 2027, as there simply aren't enough semiconductors to fill the space that is permitted.

Clearly the threat of deceleration in planned deployments in states such as Ohio and Texas is a large headache, and can cause disruption for any individual deployment. But things are moving very quickly to find LPS outside of the US, or in states that are more open to it. In 2028, our contacts are pointing to significant challenges but some confidence that those challenges can be met.

It's very early to judge the impact of midterms from a few days ago. But buyers that we talk to with sophisticated supply and demand models still tell us that we will be semiconductor constrained even through 2028. Are they right? Time will tell.

But the disconnect between Wall Street fears and Silicon Valley optimism is as wide as we have seen, and it's important to be aware of both sides of that debate. The tenor of the business is not just holding up - it is accelerating, with significant new orders coming in against fully allocated backlog every day.

So what does all this mean for stocks? To us, this all suggests a rangebound market, with the need for vibrant capital markets and resourceful management of land-power-shell constraints capping upside, but the strong tone of near-term business and many reasons for longer-term upside limiting downside as well.

If that's true, it's important not to be too negative at the bottom of the range, and we think we are closer to the bottom than the top. In particular, if capital market optimism is central to the bull case, if we use that as our guide we are likely to buy when enthusiasm is highest and sell when it is lowest, which doesn't seem to be an alpha generating strategy.

The risk of course is that this is not rangebound and that we are at the early stage of a peaking market, which would make the stocks a value trap. But we have a wide range of industry inputs suggesting duration, so we have a rangebound view.

So where would we look to be buyers as we approach a bottom? All subsectors are attractive, in our view, but there are two decision aspects to consider.

1. The "bottleneck trades" are more driven by outsized earnings power that comes with being the bottleneck - and this is where land power shell could be the biggest issue. We would include memory, obviously, but there are other categories - for example, we see Intel's earnings power as largely driven by shortages of server CPUs, given the less competitive product roadmap. We should also consider that areas such as semi cap equipment are funded through bottleneck profits (though not from increased pricing in their own business), and that in some

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ways even compute names have large prepayments that will become liabilities if memory rolls over.

Bottlenecks are the most at risk if there is a shift in bottleneck from semiconductors to non semiconductors, or from one semiconductor bottleneck to another.

That said, we have high conviction in memory sustainability even in the wake of the midterms. Our approach is less to be negative, but to be valuation sensitive. Specifically, we don't like paying a multiple on peak-ish earnings for memory; we want the current cumulative earnings power from this cycle to be the majority of market cap. That's clearly the case now, unless we are wrong and conditions erode.

2. There are a wide range of valuations and expectations priced in. Networking outlooks are very strong, but valuations are quite a bit higher, which gives us preference for compute.

With that backdrop, our subsector preference is:

- 1. GPU/XPU compute, particularly NVDA and AVGO
- 2. Memory, both SNDK and MU
- 3. Semicap (preference LRCX, MKSI, AEIS)
- 4. Analog highlighting ADI and NXPI
- 5. Networking - not negative just more expensive than peers; SITM, ALAB, SMTC, MRVL still all have positive risk rewards
- 6. Microprocessors, where near-term fundamentals are exceptional, but shortages are likely to ease well before memory, and stocks are crowded; still, we like AMD more than INTC

Details:

1) Compute is our preferred subsector with demand that remains very strong, fabless models limiting margin compression if demand weakens, and attractive multiples (NVDA/AVGO both ~9x CY28 P/E). Within compute the secondary affects of a lower than planned GW build in 2028 are part of the relative preference for our Top Pick NVDA given (1) the larger customer base relative to AMD/ASIC gives them more flexibility to go to where the power is; with sovereign business creation as a result of countries that want to use their land/power/shell to create businesses rather then let them be used by US based hyperscalers. It also affords NVIDIA significant opportunities to drive revenue sharing agreements and valuable software/services revenue streams; (2) increasing revenue per GW can drive growth even in a GW constrained environment; next gen architecture Feynman is expected to drive revenue per GW from ~$40bn with Rubin to >$50bn; (3) the increased importance on tokens per watt, where Nvidia has strong leadership. See: Reinstating Top Pick as NDR supports our positive views.

NVDA is our top pick, and we remain optimistic; that said, the large market cap tends to cap the multiple, and it behaves as a very defensive name; we think that it's important to play offense - where the memory stocks are starting to look more appealing. We do see NVIDIA as defensive, especially if bottlenecks shift from semis to land/power/shell, which is very favorable for NVDA, but in a rebound we could see better returns from memory.

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2) Memory risk reward starting to look very favorable even with land/power/shell risks For us, the multiple compression in memory is increasingly offering very compelling upside relative to those risks, and FCF earned during the upturn is being predominantly used to buy back large quantities of stock. As the stocks fall, those buybacks become even more accretive to EPS. We would highlight Sandisk, where at the current share price of $835 and buybacks in the range of 50% of revenue, they could repurchase >50% of the company between now and the end of 2028. While it's possible that land/power/shell becomes a significant enough bottleneck, we expect a resourceful industry to adapt, and we are still seeing buyers willing to put down large deposits for memory several years from now, which implies that they at least believe it's durable. We like to see most of valuation to be based on the current cycle, rather than paying a multiple on frothy numbers - we think that as stocks come down we are at that view. We like both Sandisk and Micron roughly equally at current levels.

3) Semiconductor production equipment (SPE) – Shareholder Returns to be key theme in 2027. We acknowledge the uncertainty in the data center build out and GW deployment outlook, but one thing we are confident on is that absent cuts to 2027 WFE forecasts, companies should exit the year with meaningfully stronger balance sheets or having repurchased a significant portion of their shares outstanding. LAM's newly announced $25bn buyback authorization, combined with the remaining balance at end-September, brings total authorized capacity to approximately $28bn, or 11% of market cap. AMAT and KLA each have remaining authorizations equivalent to approximately 5% of market cap, followed by AEIS, MKSI and NVMI at approximately 2–4%, and TER and ONTO below 2%. As we highlighted in our earnings report, MKS will come out with a meaningfully healthier balance sheet, with the stock trading at approx. 10% CY27 FCF.

4) Analog stocks have held up the best during the slowdown, but represent a less attractive risk reward as a group than memory or compute, in our view, given analog stocks have more limited multiple expansion potential if sentiment improves, are increasingly seeing supply constraints limit growth, and where there is better operating leverage its increasingly AI driven in nature. That leaves us with a preference for more direct exposure at lower prices.

That said, our view is that the analog cycle is continuing to strengthen, with foundry trailing edge utilization approaching 90% - while data center is less than 15% of analog demand, it's strong enough that we can start anticipating real shortages 12 months from now. Utilizations are lower at TXN, and in China, but that supply is not fungible.

We like NXPI in this space. At least in part this is because we want our analog exposure to not be overly tied to the AI trade; while we are bullish on AI, we have seen how fragile sentiment can be. NXPI offers real economy exposure at a discounted valuation, where the consensus positioning is somewhat negative given mixed automotive sell through.

We also like ADI, which is much more tied to the AI buildout given direct AI exposure, but also has strong exposure through communications infrastructure/ optical, industrial through test equipment, and real economy exposure to a strong

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inventory cycle.

5) Networking very positive, but also has priced in the most optimism. We see very strong conditions across the networking space, but stocks are generally more crowded based on our incoming call volume, and most companies that have guided to 2030 earnings power (both Marvell and Semtech), which in our view tends to provide a tough backdrop for semis stocks. In a rebound, those crowds could work for the stocks as bulls re-gross, but we just see more of a rebound possible elsewhere.

Our forced ranking is likely to offend someone, and we would note that networking risk reward still seems quite positive.

6) CPU. Our most out of consensus caution is probably in this space, and ranking it last is at least in part due to that enthusiasm. We think that CPU demand is quite good, and remains in shortage for a while - but it is the duration of that shortage that seems lower than memory, with valuations that are higher.

Why is demand good? For all the reasons that we have written about: that CPUs still play a very important role in the AI data center, but also have incremental agentic responsibilities. With that said, making the case that CPU will outgrow GPU in the AI rack - which we would agree with - is not the strong demand driver that people think it is given that a large portion of the current CPU market grows much more slowly and is not directly driven by AI. But we would certainly agree that there are multiple 50%+ years for CPU.

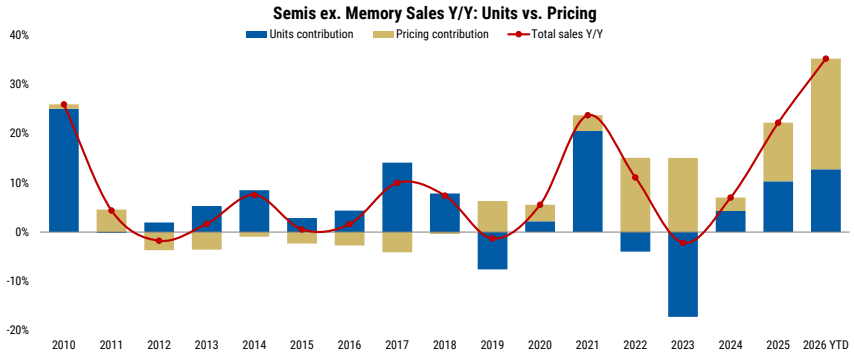
But the current shortage is at least in part driven by supply issues, with Intel only delivering 4% unit growth YTD (with over 50% price increases, that are driven by higher core counts but also higher prices). And supply growth looking forward is going to be exceptionally high, with AMD guiding (conservatively, in our view) for 100% growth next year, almost everyone in the ARM camp growing at similar rates including NVIDIA VERA but also custom ARM for hyperscalers), and Intel will likely see a large unit rebound as we get past shortages. With NAND supply growing 20%,and CPU growth at 3-4X that level, it seems highly likely that we will surpass CPU bottlenecks before other areas.

This keeps us on the sidelines for Intel, which has been the biggest beneficiary of like for like improvement. AMD is certainly in a better position in CPU, in our view, as an end to shortages would not be as impactful. Further, AMD is also in category 1, which to us is key - given how much of AMD's recent alpha has been CPU driven, we think the burden is much more on mi455 than CPU from here.

Again, forced ranking requires us to be disciplined here - and a rising tide will lift all boats. But our preference is elsewhere.

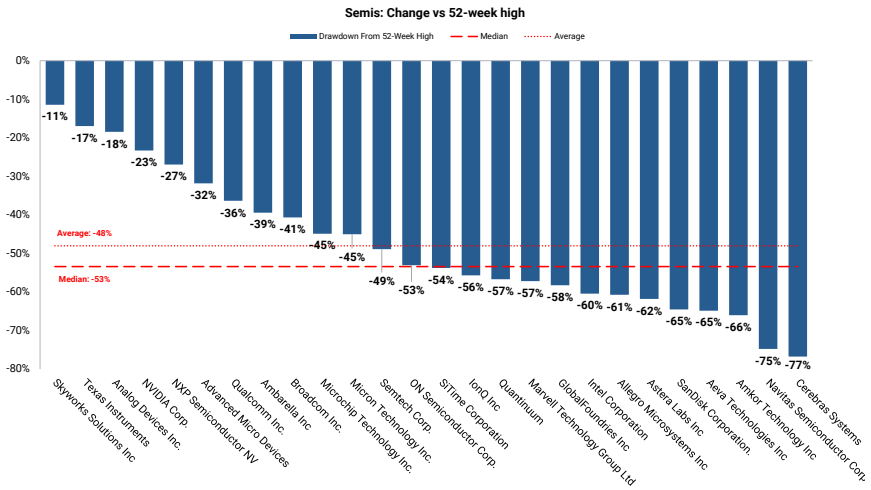
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Exhibit 1: Two thirds of semis growth in 2026 has come from price, but unit growth is also tracking to a 5yr high



Source: SIA, Morgan Stanley

Exhibit 2: The median stock in our coverage is down 53% from peak



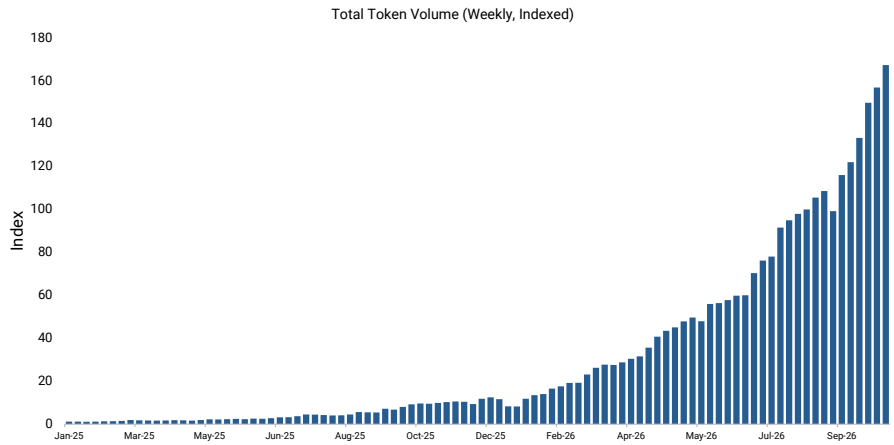
Source: Factset, Morgan Stanley Research, *prices as of 11/6 market close*

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More Picture Less Words: Compute Demand Indicators Remain Strong

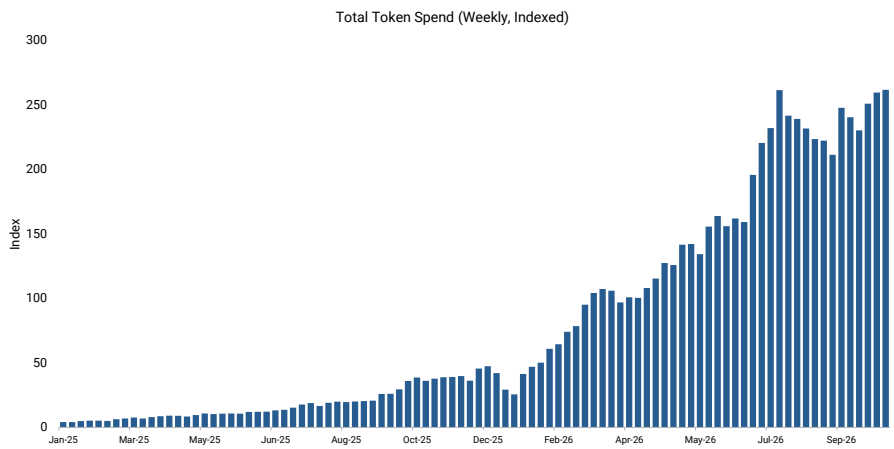
More Pictures Less Words: Compute Demand Indicators Remain Strong

Exhibit 3: Total token volume growth continues to be steep



Source: Ramp, Morgan Stanley

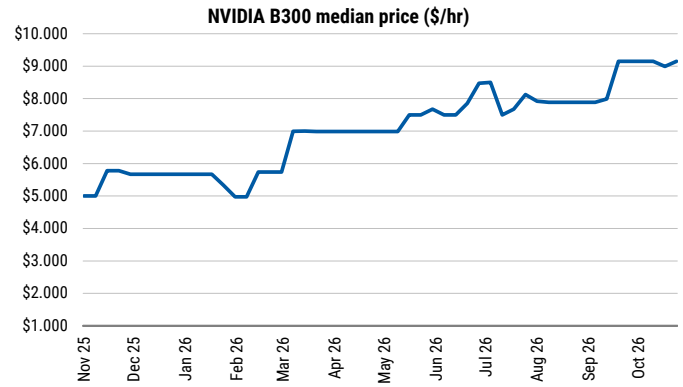
Exhibit 4: Total token spend has accelerated in recent months, returning to all time highs



Source: Ramp, Morgan Stanley

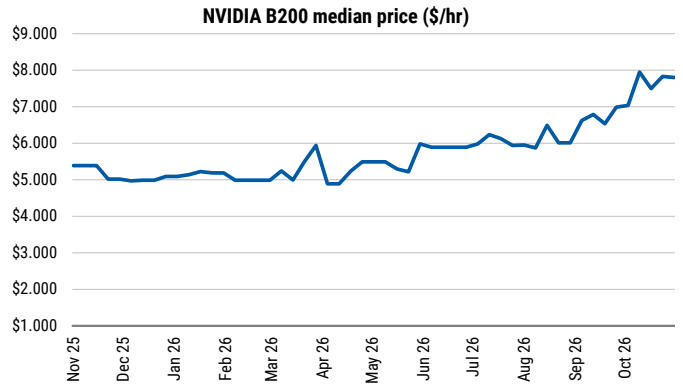
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Exhibit 5: B300 rental price



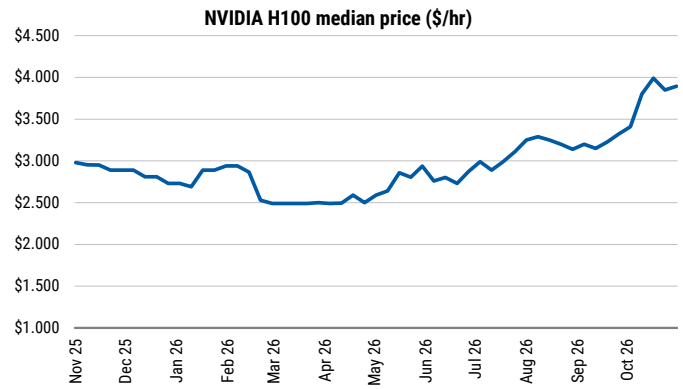
Source: getdeploying.com, Morgan Stanley

Exhibit 6: B200 rental price



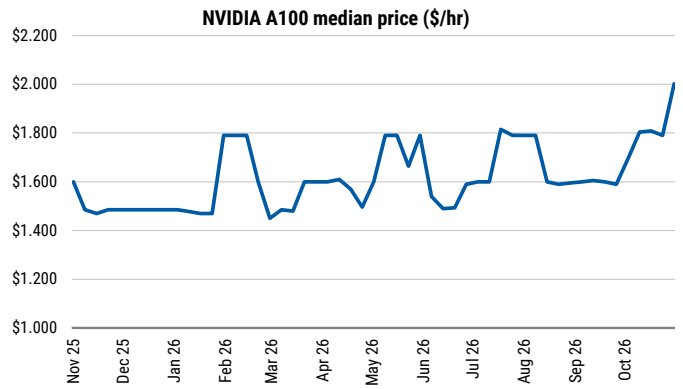
Source: getdeploying.com, Morgan Stanley

Exhibit 7: H100 rental price



Source: getdeploying.com, Morgan Stanley

Exhibit 8: A100 rental price



Source: getdeploying.com, Morgan Stanley

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(as of October 31, 2026)

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Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1575	43%	460	49%	29%	742	44%
Equal-weight/Hold	1537	42%	393	42%	26%	720	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	92	10%	16%	218	13%
Total	3,684		946			1681	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
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Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

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INDUSTRY COVERAGE: Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (11/06/2026)
Ella Tulchinsky		
Infleqtion, Inc. (INFQ.N)	E (10/15/2026)	$8.89
Xanadu Quantum Technologies Limited (XNDU.O)	E (10/15/2026)	$2.67
Joseph Moore		
Advanced Micro Devices (AMD.O)	E (06/09/2024)	$449.04
Aeva Technologies Inc (AEVA.O)	E (07/19/2021)	$11.00
Allegro Microsystems Inc (ALGM.O)	O (02/13/2026)	$28.17
Ambarella Inc (AMBA.O)	O (03/29/2016)	$58.55
Amkor Technology Inc (AMKR.O)	E (11/08/2023)	$32.82

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Analog Devices Inc. (ADI.O)	O (11/16/2023)	$363.56
Astera Labs Inc (ALAB.O)	O (05/11/2025)	$190.71
Broadcom Inc. (AVGO.O)	O (06/09/2024)	$293.69
Cerebras Systems (CBRS.O)	O (06/08/2026)	$89.68
GlobalFoundries Inc (GFS.O)	E (10/28/2024)	$38.62
Intel Corporation (INTC.O)	E (02/22/2023)	$56.28
IonQ Inc (IONQ.N)	E (04/25/2023)	$32.62
Marvell Technology Group Ltd (MRVL.O)	E (09/14/2015)	$141.20
Microchip Technology Inc. (MCHP.O)	E (07/10/2024)	$58.35
Micron Technology Inc. (MU.O)	O (10/06/2025)	$690.00
Navitas Semiconductor Corp (NVTS.O)	U (04/06/2025)	$8.61
NVIDIA Corp. (NVDA.O)	O (03/16/2023)	$186.64
NXP Semiconductor NV (NXPI.O)	O (02/11/2025)	$248.29
ON Semiconductor Corp. (ON.O)	++	$63.31
Qualcomm Inc. (QCOM.O)	E (06/24/2026)	$165.41
Quantinuum (QNT.O)	O (10/15/2026)	$37.57
SanDisk Corporation. (SNDK.O)	O (03/03/2025)	$835.06
Semtech Corp. (SMTC.O)	E (04/06/2025)	$103.21
Silicon Laboratories Inc. (SLAB.O)	E (01/19/2021)	$221.50
SiTime Corporation (SITM.O)	O (09/16/2026)	$416.92
Skyworks Solutions Inc (SWKS.O)	E (11/28/2018)	$82.19
Texas Instruments (TXN.O)	U (04/13/2020)	$277.36
Wolfspeed, INC (WOLF.N)	NR (04/06/2025)	$23.79
Lee Simpson		
Arm Holdings plc (ARM.O)	E (04/07/2026)	$174.62
Cadence Design Systems Inc (CDNS.O)	O (02/14/2024)	$319.92
PDF Solutions Inc (PDFS.O)	O (09/08/2026)	$37.61
Synopsys Inc. (SNPS.O)	O (09/08/2026)	$450.46

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

INDUSTRY COVERAGE: Semiconductor Capital Equipment

COMPANY (TICKER)	RATING (AS OF)	PRICE* (11/06/2026)
Shane Brett		
Advanced Energy Industries Inc. (AEIS.O)	O (07/22/2026)	$197.01
Applied Materials Inc. (AMAT.O)	E (05/18/2026)	$332.72
Camtek (CAMT.O)	U (11/02/2026)	$97.05
KLA Corp (KLAC.O)	O (01/15/2026)	$138.49
Lam Research Corp (LRCX.O)	O (05/18/2026)	$200.44
MKS Inc. (MKSI.O)	O (08/04/2024)	$192.27
Nova Ltd (NVMI.O)	E (12/01/2025)	$313.18
ONTO Innovation Inc (ONTO.N)	O (06/14/2026)	$173.28
Teradyne Inc (TER.O)	E (07/30/2025)	$218.22

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